

**FILE : CH8EG1**

```
num=int(input("enter an integer:"))
if num%2 ==0:
    print(num,"is even number")
else:
    print(num,"is odd number")
```

**FILE : CH8EG2**

```
x=y=z=0
x=float(input("enter first number:"))
y=float(input("enter second number:"))
z=float(input("enter third number:"))
max=x
if y>max:
    max=y
if z>max:
    max=z
print("Largest number is ",max)
```

**FILE : CH8EG3**

```
sum1=sum2=0
num1=int(input("enter number1:"))
num2=int(input("enter number2:"))
num3=int(input("enter number3:"))
sum1=num1+num2+num3
if num1==num2:
    if num3!=num1:
        sum2+=sum3
else:
    if num1==num3:
        sum2+=num2
    else:
        if num1==num3:
            sum2+=num2
        else:
            if num2==num3:
                sum2+=num1
            else:
                sum2+=num1+num2+num3
```

```

print("number are",num1,num2,num3)
print("sum of three given number is",sum1)
print("sum of non duplicated number is",sum2)
print()
print("OR")
print()
num1=int(input("enter number1:"))
num2=int(input("enter number2:"))
num3=int(input("enter number3:"))
sum1=num1+num2+num3
sum2=0
if num1!= num2 and num1!= num3:
    sum2+=num1
if num2!= num1 and num2!= num3:
    sum2+=num2
if num3!= num1 and num3!= num2:
    sum2+=num3
print("number are",num1,num2,num3)
print("sum of three given number is",sum1)
print("sum of non duplicated number is",sum2)
print()

```

#### FILE : CH8EG4

```

num1=int(input(" enter the dividend "))
num2=int(input(" enter the divisor "))
if num1%num2==0:
    print(num1, " is divisible by ", num2 )
else:
    print(num1, " is not divisible by ", num2 )

```

#### FILE : CH8EG5

```

print("enter five number below")
num1=float(input("first number:"))
num2=float(input("second number:"))
num3=float(input("third number:"))
num4=float(input("fourth number:"))
num5=float(input("fifth number:"))
divisor=float(input("enter the divisor number:"))

```

```

count=0
print("multiple of", divisor,"are:")
remainder=num1%divisor
if remainder==0:
    print(num1,sep=" ")
    count+=1
remainder=num2%divisor
if remainder==0:
    print(num2,sep=" ")
    count+=1
remainder=num3%divisor
if remainder==0:
    print(num3,sep=" ")
    count+=1
remainder=num4%divisor
if remainder==0:
    print(num4,sep=" ")
    count+=1
remainder=num5%divisor
if remainder==0:
    print(num5,sep=" ")
    count+=1
print()
print(count,"multiples of",divisor,"found")

```

**FILE : CH8EG6**

**# menu driven program**

```

radius=float(input("enter radius of the circle :"))
print("1.calculate Area")          # menu1
print("2.calculate perimeter(circumference)") # menu2
c=int (input("enter your choise(1 or 2):")) # menu choice
if c==1:
    area=3.14159*radius*radius
    print("area of circle with radius",radius,'is',area)
elif c==2:
    perm=2*314159*radius
    print("circumference of circle with radius",radius,'is',perm)
else:
    print("Wrong choice entered")

```

**FILE : CH8EG7**

```
sale=int(input("enter the amount of monthly sell: "))
if sale>=30000:
    discount=sale*0.18
elif sale>=20000:
    discount=sale*0.15
elif sale>=10000:
    discount=sale*0.10
else:
    discount=sale*0.05
print("discount sale:",discount)
```

**FILE : CH8EG8**

```
x=int(input("enter first number:"))
y=int(input("enter second number:"))
z=int(input("enter third number:"))
min=mid=max=None
if x<y and x<z:
    if y<z:
        min,mid,max=x,y,z
    else:
        min,mid,max=x,z,y
elif y<x and y<z:
    if x<z:
        min,mid,max=y,x,z
    else:
        min,mid,max=y,z,x
else:
    if x<y:
        min,mid,max=z,x,y
    else:
        min,mid,max=z,y,x
print("number in ascding order:",min,mid,max)
```

**FILE : CH8EG9**

```
ch=input("enter a single character: ")
if ch>='A' and ch<='Z':
    print(" It is a capital letter")
elif ch>='a' and ch<='z':
    print(" It is a small letter")
elif ch>='0' and ch<='9':
    print(" It is a digit")
elif ch==' ':
    print(" It is a space ")
else:
    print("It is a special character")
```

**FILE : CH8EG10**

```
import math
a=int(input("Enter a: "))
b=int(input("Enter b: "))
c=int(input("Enter c: "))
if a==0:
    print("value of a",a,'should not be zero')
    print("\n Aborting!!!")
else:
    delta=b*b-4*a*c
    if delta>0:
        root1=(-b+math.sqrt(delta))/(2*a)
        root2=(-b-math.sqrt(delta))/(2*a)
        print("Roots are REAL and UNEQUAL")
        print("Root1=",root1,",Root2=",root2)
    elif delta==0:
        root1=-b/(2*a)
        print("Roots are REAL and UNEQUAL")
        print("Root1=",root1,",Root2=",root1)
    else:
        print("Root are COMPLEX AND IMAGINARY")
```

**FILE : CH8EG11**

```
d=int(input("Enter deposit: "))
t=int(input("Enter time: "))
    # NAMING/assigning OF CONDITIONS to variables
ef5p= d<2000 and t>=2
    # when deposit is less than Rs. 2000 and for 2 years or more
    at 5%/a
ef7p= 2000<=d <=6000 and t>=2
    # when deposit is Rs. 2000 or more than but less than equal
    to 6000 and for 2 years or more at 7%/a
ef8p= d>= 6000 and t>=1
    # when deposit is more than Rs. 6000 and for 1 years or more
    at 8%/a
ef10p= t>=5
    # when all deposit for 5 years or more at 10%/a
if ef5p:          # means ef5p == True
    r=5
elif ef7p:        # means ef7p == True
    r=7
elif ef8p:        # means ef8p == True
    r=8
elif ef10p:       # means ef10p == True
    r=10
else:
    r=0
si=d*t*r/100
print("si = ",si)
```

**FILE : CH8EG12**

```
print(3 in [1,2,3,4])
print(5 in [1,2,3,4])
print(5 not in [1,2,3,4])
print('a'in "trade")
print('ash'in "trash")
```

#### FILE : CH8EG13

```
#1
print()
print("#1")
l=int(input ("Enter the lower limit "))
u=int(input ("Enter the upper limit "))
s=int(input ("Enter the step value "))
for a in range(l,u,s):      # list given by range
    print(a,end=" ")
print()
```

```
#2
print()
print("#2")
for a in range(10):        # list given by range
    print(a,end=" ")
print()
```

#### FILE : CH8EG14

```
#1
print()
print("#1")
for a in [1,4,7]:          # list
    print(a,end=",")
    print(a*a,end=" \t")
```

```
#2
print();print("#2")
for a in range(1,5):       # list given by range
    print(a,end=",")
    print(a*a,end=" \t")
```

```
#3
print();print("#3")
for a in range(5,1,-1):    # list given by range
    print(a,end=",")
    print(a*a,end=" \t")
```

#4

```
print()
print("#4")
for a in range(12,18):      # list given by range
    print(a,end=",")
    print(a*a,end=" \t")
print()
```

#5

```
print()
print("#5")
for ch in 'calm':
    print(ch,end=",")
```

FILE : CH8EG15

```
num=int(input("enter a value "))
for a in range (1,21):
    print(num,'x',a,'=',num*a)
```

FILE : CH8EG16

```
sum=0
for n in range(11,101):
    sum+=n
    print("sum of natural <=",n,"is",sum)
```

FILE : CH8EG17

```
sum=0
l=int(input("enter the value1 :"))
u=int(input("enter the value2 :"))
for n in range(l,u):
    sum+=n
print("sum of natural <=",n,"is",sum)
```



**FILE : CH8EG18**

```
a=5
while a>0:
    print("hello",a)
    a =a-3
print ("loop over!!!")
```

**FILE : CH8EG19**

**#1**

```
print()
print("#1")
n=1
while n<5:
    print("square of",n,'is',n*n)
    n+=1
print("Thank you")
```

**#2**

```
print()
print("#2")
n=5
while n>1:
    print("square of",n,'is',n*n)
    n-=1
print("Thank you")
```

**FILE : CH8EG20**

```
n=int(input("enter a number:"))
a=1;fact=1
while a<=n:
    fact*=a
    a+=1
print("the factorial of",n,"is",fact)
```

**FILE : CH8EG21**

```
n=int(input("up to which natural number? "))
ctr=1;sum_even=sum_odd=0
while ctr<=n:
    if ctr%2==0:
        sum_even+=ctr
    else:
        sum_odd+=ctr
    ctr+=1
print("the sum of even integers is",sum_even)
print("the sum of odd integers is",sum_odd)
```

**FILE : CH8EG22**

```
a=b=c=0
for i in range(1,5):
    a=int(input("enter the number1 :"))
    b=int(input("enter the number2 :"))
    if b==0:
        print("divisor by zero error! abortion! ")
        break
    else:
        c=a//b
        print("Question=",c)
print("program over! ")
```

**FILE : CH8EG23****# gussing game**

```
import random
n=random.randint(10,50)
ctr=0
print("guess a number in range 10..50 :")
while ctr<5:
    print("trial number : ",ctr+1)
    guess= int(input("guess your number :"))
    if guess==n:
        print("you win!!:)",n)
        break
```

```

elif n-5<guess and guess<n+5 :
    print("you are near ")
else:
    print("wild guess ")
    ctr+=1
if not ctr <5:      # OR ctr>=5:  'this if is part of while loop
    print("you lose:(\n the number was",n)

```

FILE : CH8EG24

#1

```

print()
print("#1")
a=2
while True:      #while True;      (boolean literal) creates an
infinite loop
    print(a)
    a*=2
    if a>100:    # to break the infinite looping
        break

```

#2

```

print()
print("#2")
for i in range(1,5):
    print("enter two numbers")
    a=int(input("enter the number 1 :"))
    b=int(input("enter the number 2 :"))
    if b==0:
        print("\n the denominator cannot be zero . enter again!")
        continue
    c=a//b
    print("Question=",c)

```

**FILE : CH8EG25**

```
print("the loop with 'break' produce output as:")
for i in range(1,11):
    if i%3==0:
        break
    else:
        print(i)
print("the loop'continue'produces output as:")
for i in range (1,11):
    if i%3==0:
        continue
    else:
        print(i)
```

**FILE : CH8EG26**

#1

```
print("#1")
for a in range(1,4):
    if a%2==0:
        break
    print("Element is ",end=' ')
    print(a)
else:
    print("ending loop after printing all elements of sequence")
```

#2

```
print();print("#2")
for a in range(1,4):
    print("Element is ",end=' ')
    print(a)
    if a%2==0:
        break
    else:
        print("ending loop after printing all elements of
sequence")
```

#3

```
print("#3")
count=sum=0
ans='y'
while ans=='y':
    num=int(input("enter the number :"))
    if num<0:
        print("number entered is below zero. Aborting!")
        break
    sum=sum+num
    count=count+1
    ans=input("want to enter more number? (y/n)")
else:
    print("you entered",count,"number so far")
print("sum of number entered is",sum)
```

FILE : CH8EG27

```
num=int(input("enter number : "))
lim=int(num/2)+1
for i in range(2,lim):
    rem=num%i
    if rem==0:
        print(num,"is not a prime number ")
        break
    else:
        print(num,"is a prime number")
```

FILE : CH8EG28

#1

```
print()
print("#1")
while(3>4):                # 3<4 it will create infinite loop
    print("in loop")
else:                       # 3>4 it will not execute the loop even
once
    print("exiting from while loop")
```

#2

```
print()
print("#2")
for a in range (10,1,-1):
    print("in loop",a)
else:
    # else works after normal termination of
    for loop
    print("exiting from for loop")
```

#3

```
print()
print("#3")
for i in range (1,6):
    for j in range (1,i):
        print("*",end=" ")
    print()
```

#4

```
print()
print("#4")
for outer in range (5,10,4):
    for inner in range (1,outer,2):
        print(outer, inner)
    print()
```

#5

```
print()
print("#5")
for a in range (3):
    for b in range (5,7):
        print("for a=",a,"b is now",b)
    print()
```

FILE : CH8EG29

#1

```
print("#1")
for a in range (3):
    for b in range (5,7):
        print ("*",end='')
    print()
```

#2

```
print("#2")
for num in range (15,25):
    for i in range (2,num):
        if num%i==0:
            j=num/i
            print("founded a factori(",i,")for",num)
            break
    else:
        print(num,"is a prime number")
```